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END TERM EXAMINATION

FIRST SEMESTER [B.TECH] DECEMBER 2024 - JANUARY 2025

Paper Code: BS-105**Subject: Applied Physics-I****Time: 3 Hours****Maximum Marks: 60**

Note: Attempt five questions in all including Q. No. 1 which is compulsory. Select one question from each unit. Assume missing data, if any.

Q. No.	Question	Marks
Q.1	Attempt all of the following:	(5 × 4 = 20)
(a)	What are inertial and non-inertial frames of reference? Is an aircraft in steady flight an inertial frame or non-inertial frame?	(4)
(b)	Give the physical significance of divergence and curl of a field.	(4)
(c)	What is optic axis and principal section of a doubly refracting crystal?	(4)
(d)	Why population inversion is necessary for laser action?	(4)
(e)	Calculate the change in entropy when 2 gm of ice melts into water at the same temperature. Latent heat of ice = 80 Cal/gm.	(4)
UNIT – I		
Q.2	Write down the zeroth law of thermodynamics. Explain how it introduces the temperature of a system as a function of state.	(5)
(a)	What do you understand by indicator diagram? Draw indicator diagram for isochoric and isobaric processes.	(3)
(b)	What is Carnot's theorem? Give its significance.	(2)
OR		
Q.3	Define the term 'Entropy'. Derive an expression for change of entropy for the following processes: 1. Isochoric process 2. Isobaric process 3. Isothermal process 4. Adiabatic process	(7)
(b)	A cyclic heat engine operates between a source temperature of 800 °C and a sink temperature of 30 °C. What is the least rate of heat rejection per kW net output of the engine?	(3)
UNIT – II		
Q.4	Discuss in brief the inconsistencies in the Ampere's law and describe how Maxwell fixed up this. Further discuss in brief the characteristics of transient and displacement current.	(5)
(b)	Write down the three dimensional wave equation and obtain its solution.	(5)
OR		
Q.5	Derive the equation for Electric and Magnetic field intensities for an electromagnetic wave propagating in the z-direction in a charge and current free region. Find the following: (i) Refractive index (ii) Poynting vector (iii) Intrinsic impedance	(7)
(b)	State and deduce Poynting theorem for the flow of energy in an electromagnetic field.	(3)

Q. No.	Question	Marks
UNIT — III		
Q.6 (a)	Give the construction and theory of plane transmission grating. Explain the formation of spectra by it.	(4)
(b)	Show that the radius for n^{th} dark ring formed in a Newton's rings experiment is proportional to $\sqrt{\lambda}$.	(4)
(c)	What are coherent sources?	(2)
OR		
Q.7 (a)	Describe construction and working of a Nicol prism.	(3)
(b)	What do you mean by optical activity? Give Fresnel's theory of optical activity and derive the necessary expression for the optical rotation.	(3)
(c)	Find the thickness of a quarter wave plate of quartz for light of wavelength 5893 Å. The refractive indices for ordinary and extraordinary rays are 1.544 and 1.553 respectively.	(4)
UNIT — IV		
Q.8 (a)	State the fundamental postulates of special theory of relativity and deduce the Lorentz transformation equations from them and discuss how these equations account for the phenomenon of length contraction.	(8)
(b)	Derive an expression for time dilation. A clock measures the proper time. With what speed it should move relative to an observer so that it appears to go slow by 30s in 24 hours.	(4)
OR		
Q.9 (a)	What do you understand by 3 and 4 levels LASER? What are the advantages of 3 level over 4 level LASER?	(2)
(b)	Describe the construction and working of a Ruby laser with the help of a well labeled diagram.	(5)
(c)	Describe Einstein coefficients for laser action.	(3)

**** End of Question Paper ****